

MEng (Engineering) Examination 2020

Year 1

77/100

AERO40006 Introduction to Mathematics

Thursday 15th October 2020: 11.00 to 12.00
[1 hour examination + max 30 minutes upload time]

OPEN BOOK EXAMINATION

There are FIVE questions. All questions carry equal weight.
Candidates may obtain full marks for complete answers to **ALL** questions.

A datasheet is provided.

This is an open book assessment. We have worked hard to create exams that assesses synthesis of knowledge rather than factual recall. Thus, access to the internet, notes or other sources of factual information in the time provided might not be helpful and may well limit your time to successfully synthesise the answers required.

Where individual questions rely more on factual recall and may therefore be less discriminatory in an open book context, we may compare the performance on these questions to similar style questions in previous years and we may scale or ignore the marks associated with such questions or parts of the questions.

The use of the work of another student, past or present, constitutes plagiarism. Giving your work to another student to use may also constitute an offence. Collusion is a form of plagiarism and will be treated in a similar manner. This is an individual assessment and thus should be completed solely by you. The College will investigate all instances where an examination or assessment offence is reported or suspected, using plagiarism software, vivas and other tools, and apply appropriate penalties to students. In all examinations we will analyse exam performance against previous performance and against data from previous years and use an evidence-based approach to maintain a fair and robust examination. As with all exams, the best strategy is to read the question carefully and answer as fully as possible, taking account of the time and number of marks available

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Section A

Question 1

Part (a)

Calculate the derivative of $y(x) = 5^x$.

50 [50%]

Part (b)

Find dy/dx when $y(x)$ is implicitly given by $y = x^2 + \sin(xy)$.

50 [50%]

$$(a) \ln[y(x)] = \ln 5^x = x \ln 5 \quad \therefore \frac{1}{y} \frac{dy}{dx} = \ln 5$$

$$\therefore \frac{dy}{dx} = y \ln 5 = 5^x \ln 5$$

$$(b) \frac{dy}{dx} = 2x + y \cos(xy) + x \frac{dy}{dx} \cos(xy)$$

$$\therefore (1 - x \cos(xy)) \frac{dy}{dx} = 2x + y \cos(xy)$$

$$\therefore \frac{dy}{dx} = \frac{2x + y \cos(xy)}{1 - x \cos(xy)}$$

Question 2

Part (a)

Evaluate the integral $\int x \ln(x) dx$.

50 [50%]

Part (b)

The Fourier coefficient, a_n , for a triangular waveform is given by

$$a_n = \frac{1}{\pi^2 n^2} \int_0^{2\pi} (\omega t) \cos(n\omega t) d(\omega t)$$

where n is a positive integer. Find a_n .

[50%]

(a) For $\int \ln(x) dx$, $\int \ln(x) dx = x \ln(x) - x + C$

$$\therefore \int x \ln(x) dx = x^2 \ln(x) - x^2 - \int x \ln(x) - x dx + C$$

$$= x^2 \ln(x) - x^2 - \int x \ln(x) dx + \frac{1}{2} x^2 + C$$

$$\therefore 2 \int x \ln(x) dx = x^2 \ln(x) - \frac{1}{2} x^2 + C$$

$$\therefore \int x \ln(x) dx = \frac{1}{2} x^2 \ln(x) - \frac{1}{4} x^2 + C$$

D	I
+	$x \ln(x)$
-	$x \ln(x) - x$
+	0
	...

Question 2

Part (a)

Evaluate the integral $\int x \ln(x) dx$.

[50%]

Part (b)

The Fourier coefficient, a_n , for a triangular waveform is given by

$$a_n = \frac{1}{\pi^2 n^2} \int_0^{2\pi} (\omega t) \cos(n\omega t) d(\omega t)$$

where n is a positive integer. Find a_n .

45 [50%]

$$(b) \quad a_n = \frac{1}{\pi^2 n^2} \int_0^{2\pi} (\omega t) \cos(n\omega t) d(\omega t)$$

$$\text{Let } \omega t = x$$

$$\therefore \int_0^{2\pi} (\omega t) \cos(n\omega t) d(\omega t) = \int_0^{2\pi} x \cos(nx) dx$$

$$= \left[\frac{x}{n} \sin(nx) + \frac{1}{n^2} \cos(nx) \right]_0^{2\pi}$$

$$= \frac{2\pi}{n} \sin(2n\pi) + \frac{1}{n^2} \cos(2n\pi) + \frac{1}{n^2}$$

$$- \frac{1}{n^2}$$

$$= \frac{2}{n^2} = 0$$

$$\therefore a_n = \frac{1}{\pi^2 n^2} \frac{2}{n^2} = \frac{2}{\pi^2 n^4} = 0$$

	D	I
+	x	cos(nx)
-	1	$\frac{1}{n} \sin(nx)$
+	0	$-\frac{1}{n^2} \cos(nx)$

Question 3

Express $I_n = \int (x^2 + 3)^n dx$ as a function of I_{n-1} , n and x .

0 [100%]

$$I_{n-1} = \int (x^2 + 3)^{n-1} dx = \frac{1}{2n} (x^2 + 3)^n + C$$

$$I_n = \int (x^2 + 3)^n dx = \frac{1}{2(n+1)x} (x^2 + 3)^{n+1} + C$$

$$\therefore \frac{I_n}{I_{n-1}} = \frac{(2nx)(x^2 + 3)^{n+1}}{2(n+1)x(x^2 + 3)^n} = \frac{n}{n+1} (x^2 + 3)$$

$$\therefore I_n = \frac{n}{n+1} (x^2 + 3) I_{n-1}$$

$$I_n = x(x^2 + 3)^n - \int 2nx(x^2 + 3)^{n-1} dx$$

$$= x(x^2 + 3)^n - 2n \int (x^2 + 3 - 3)(x^2 + 3)^{n-1} dx$$

$$= x(x^2 + 3)^n - 2n \int (x^2 + 3)^n dx + 6n \int (x^2 + 3)^{n-1} dx$$

$$= x(x^2 + 3)^n - 2n I_n + 6n I_{n-1}$$

$$\therefore (2n+1) I_n = x(x^2 + 3)^n + 6n I_{n-1}$$

$$\therefore I_n = \frac{x(x^2 + 3)^n}{2n+1} + \frac{6n}{2n+1} I_{n-1}$$

	D	I
+	$(x^2 + 3)^n$	1
-	$2nx(x^2 + 3)^{n-1}$	x

Section B

Question 4

Part (a)

Split the expression $\frac{x+5}{x^2-2x+1}$ into partial fractions.

50 [50%]

Part (b)

For which x the series $\sum_{n \geq 1} \frac{(x-2)^n}{n}$ converge?

40 [50%]

$$(a) \frac{x+5}{x^2-2x+1} = \frac{A}{(x-1)^2} + \frac{B}{x-1}$$

$$\therefore A + B(x-1) = x+5$$

$$\therefore Bx + (A-B) = x+5 \quad \therefore B=1, A=6$$

$$\therefore \frac{x+5}{x^2-2x+1} = \frac{6}{(x-1)^2} + \frac{1}{x-1}$$

$$(b) \lim_{n \rightarrow \infty} \frac{\left| \frac{(x-2)^{n+1}}{n+1} \right|}{\left| \frac{(x-2)^n}{n} \right|} = \lim_{n \rightarrow \infty} \left| \frac{n}{n+1} (x-2) \right| = |x-2|$$

If $|x-2| < 1$, series converge

$$-1 < x-2 < 1$$

$$\therefore 1 < x < 3$$

... If $|x-2| = 1$, $x = \pm 3$

$$\therefore \text{for } x=3, \sum_{n=1}^{\infty} \frac{(x-2)^n}{n} = \sum_{n=1}^{\infty} \frac{1}{n} \longrightarrow \text{diverge}$$

$$\text{for } x=1, \sum_{n=1}^{\infty} \frac{(x-2)^n}{n} = \sum_{n=1}^{\infty} \frac{(-1)^n}{n} \longrightarrow \text{converge}$$

$$\therefore 1 \leq x < 3$$

Question 5

Part (a)

Evaluate the limit $\lim_{x \rightarrow 0} \frac{e^{\sin(x)} - 1 - x}{x^2}$.

50 [50%]

Part (b)

Evaluate the limit $\lim_{x \rightarrow \infty} \frac{\ln(x)}{\sqrt{x}}$.

50 [50%]

$$\begin{aligned} \text{(a)} \quad \lim_{x \rightarrow 0} \frac{e^{\sin(x)} - 1 - x}{x^2} &= \lim_{x \rightarrow 0} \frac{\cos(x)e^{\sin(x)} - 1}{2x} \\ &= \lim_{x \rightarrow 0} \frac{-\sin(x)e^{\sin(x)} + \cos^2(x)e^{\sin(x)}}{2} = \frac{1}{2} \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad \lim_{x \rightarrow \infty} \frac{\ln x}{\sqrt{x}} &= \lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{\frac{1}{2x^{\frac{1}{2}}}} = \lim_{x \rightarrow \infty} \frac{2x^{\frac{1}{2}}}{x} = \lim_{x \rightarrow \infty} \frac{2}{x^{\frac{1}{2}}} \\ &= 0 \end{aligned}$$